SQL QUERY IMPLEMENTATION FOR E-COMMERCE DATABASE

ShopSphere – Online E-Commerce Database Management System

Objectives

1. To perform SELECT, WHERE, ORDER BY, and DISTINCT queries.
2. To search products based on price, category, and availability.
3. To retrieve customer and product information.
4. To apply filtering conditions.
5. To generate basic business reports.

1. Basic SQL Queries

1.1 SELECT Query

The SELECT statement is used to retrieve data from a table.

Display all products

```
SQL> SELECT * FROM Product;

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
CATEGORY_ID
-----------
       101 Laptop                                             45000
         101

       102 Mouse                                                500
         102

       103 Keyboard                                            1200
         102


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
CATEGORY_ID
-----------
       104 Mobile Phone                                       18000
         103

       105 Headphones                                           800
         104
```

Display selected product details

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product;

PRODUCT_ID PRODUCT_NAME                                          PRICE
---------- -------------------------------------------------- ----------
       101 Laptop                                              45000
       102 Mouse                                                 500
       103 Keyboard                                             1200
       104 Mobile Phone                                        18000
       105 Headphones                                            800
```

1.2 WHERE Clause

The WHERE clause is used to filter records based on a condition.

Find products with price less than 1000

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price < 1000;

PRODUCT_ID PRODUCT_NAME                                          PRICE
---------- -------------------------------------------------- ----------
CATEGORY_ID
-----------
       102 Mouse                                                 500
        102

       105 Headphones                                            800
        104
```

Find products with price greater than 100

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price > 100;

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
CATEGORY_ID
-----------
       101 Laptop                                              45000
        101

       102 Mouse                                                 500
        102

       103 Keyboard                                             1200
        102


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
CATEGORY_ID
-----------
       104 Mobile Phone                                        18000
        103

       105 Headphones                                            800
        104
```

1.3 ORDER BY

Products from lowest to highest price

```
SQL> SELECT *
  2  FROM Product
  3  ORDER BY Price ASC;

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
CATEGORY_ID
-----------
       102 Mouse                                                 500
        102

       105 Headphones                                            800
        104

       103 Keyboard                                             1200
        102


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
CATEGORY_ID
-----------
       104 Mobile Phone                                        18000
        103

       101 Laptop                                              45000
        101
```

Products from highest to lowest price

```
SQL> SELECT *
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID PRODUCT_NAME                                         PRICE
---------- -------------------------------------------------- ----------
CATEGORY_ID
------------
       101 Laptop                                              45000
        101

       104 Mobile Phone                                        18000
        103

       103 Keyboard                                             1200
        102


PRODUCT_ID PRODUCT_NAME                                         PRICE
---------- -------------------------------------------------- ----------
CATEGORY_ID
------------
       105 Headphones                                            800
        104

       102 Mouse                                                 500
        102
```

1.4 DISTINCT

The DISTINCT keyword is used to display unique values.

Display unique categories

```
SQL> SELECT DISTINCT Category_ID
  2  FROM Product;

CATEGORY_ID
-----------
        101
        102
        103
        104
```

2. Search Products

2.1 Search Products Based on Price

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price BETWEEN 100 AND 1000;

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
CATEGORY_ID
-----------
       102 Mouse                                                 500
        102

       105 Headphones                                            800
        104
```

This displays products whose price is between 100 and 400. If no products match this range, Oracle displays “no rows selected.”

2.2 Search Products Based on Category

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Category_ID = 101;

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
CATEGORY_ID
-----------
       101 Laptop                                                45000
        101
```

This displays products belonging to category 101.

3. Retrieve Customer Information

Display all customers

```
SQL> SELECT *
  2  FROM Customer;

CUSTOMER_ID CUSTOMER_NAME                                      GENDER
----------- -------------------------------------------------- ----------
       AGE CITY                                               PHONE_NUMBER
---------- -------------------------------------------------- ---------------
EMAIL
--------------------------------------------------------------------------------
    SALARY
----------
        101 Arun Kumar                                         Male
        25 Chennai                                             9876543210
arun@gmail.com
     30000


CUSTOMER_ID CUSTOMER_NAME                                      GENDER
----------- -------------------------------------------------- ----------
       AGE CITY                                               PHONE_NUMBER
---------- -------------------------------------------------- ---------------
EMAIL
--------------------------------------------------------------------------------
    SALARY
----------
        102 Priya Sharma                                       Female
        22 Chennai                                             9876543211
priya@gmail.com
     33000


CUSTOMER_ID CUSTOMER_NAME                                      GENDER
----------- -------------------------------------------------- ----------
       AGE CITY                                               PHONE_NUMBER
---------- -------------------------------------------------- ---------------
EMAIL
--------------------------------------------------------------------------------
    SALARY
----------
        103 Rahul Raj                                          Male
        28 Coimbatore                                          9876543212
rahul@gmail.com
     35000


CUSTOMER_ID CUSTOMER_NAME                                      GENDER
----------- -------------------------------------------------- ----------
       AGE CITY                                               PHONE_NUMBER
---------- -------------------------------------------------- ---------------
EMAIL
--------------------------------------------------------------------------------
    SALARY
----------
        104 Divya Sri                                          Female
        24 Chennai                                             9876543213
divya@gmail.com
     32000


CUSTOMER_ID CUSTOMER_NAME                                      GENDER
----------- -------------------------------------------------- ----------
       AGE CITY                                               PHONE_NUMBER
---------- -------------------------------------------------- ---------------
EMAIL
--------------------------------------------------------------------------------
    SALARY
----------
        105 Karthik S                                          Male
        26 Madurai                                             9876543214
karthik@gmail.com
     30000
```

Search for a particular customer

```
SQL> SELECT *
  2  FROM Customer
  3  WHERE Customer_ID = 101;

CUSTOMER_ID CUSTOMER_NAME                                      GENDER
----------- -------------------------------------------------- ----------
       AGE CITY                                               PHONE_NUMBER
---------- -------------------------------------------------- ---------------
EMAIL
--------------------------------------------------------------------------------
    SALARY
----------
        101 Arun Kumar                                         Male
        25 Chennai                                             9876543210
arun@gmail.com
     30000
```

4. Apply Filtering Conditions

4.1 AND Condition

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price > 500
  4  AND Price < 2000;

PRODUCT_ID PRODUCT_NAME                                          PRICE
---------- -------------------------------------------------- ----------
CATEGORY_ID
-----------
       103 Keyboard                                             1200
        102

       105 Headphones                                            800
        104
```

4.2 OR Condition

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Category_ID = 101
  4  OR Category_ID = 102;

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
CATEGORY_ID
-----------
       101 Laptop                                             45000
        101

       102 Mouse                                                500
        102

       103 Keyboard                                            1200
        102
```

4.3 IN Operator

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Category_ID IN (101, 102, 103);

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
CATEGORY_ID
-----------
       101 Laptop                                             45000
         101

       102 Mouse                                                500
         102

       103 Keyboard                                            1200
         102


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
CATEGORY_ID
-----------
       104 Mobile Phone                                       18000
         103
```

4.4 BETWEEN Operator

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price BETWEEN 1000 AND 5000;

PRODUCT_ID PRODUCT_NAME                                         PRICE
---------- -------------------------------------------------- ----------
CATEGORY_ID
-----------
       103 Keyboard                                            1200
         102
```

5. Basic Business Reports

5.1 Product Price Report

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID PRODUCT_NAME                                         PRICE
---------- -------------------------------------------------- ----------
       101 Laptop                                             45000
       104 Mobile Phone                                       18000
       103 Keyboard                                            1200
       105 Headphones                                           800
       102 Mouse                                                500
```

5.2 Customer Order Report

```
SQL> SELECT C.Customer_ID, C.Customer_Name, O.Order_ID
  2  FROM Customer C
  3  JOIN Orders O
  4  ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
        101 Arun Kumar                                           1001
        102 Priya Sharma                                         1002
        103 Rahul Raj                                            1003
        104 Divya Sri                                            1004
```

5.3 Product Order Report

```
SQL> SELECT Product_ID, Order_ID
  2  FROM Order_Details;

PRODUCT_ID   ORDER_ID
---------- ----------
       101       1001
       102       1002
       103       1003
       104       1004
```

5.4 Customer-wise Total Orders

```
SQL> SELECT C.Customer_ID, C.Customer_Name,
  2         COUNT(O.Order_ID) AS Total_Orders
  3  FROM Customer C
  4  LEFT JOIN Orders O
  5  ON C.Customer_ID = O.Customer_ID
  6  GROUP BY C.Customer_ID, C.Customer_Name;

CUSTOMER_ID CUSTOMER_NAME                                  TOTAL_ORDERS
----------- -------------------------------------------------- ------------
        101 Arun Kumar                                              1
        102 Priya Sharma                                            1
        103 Rahul Raj                                               1
        104 Divya Sri                                               1
        105 Karthik S                                               0
```

5.5 Customer-wise Total Purchase Amount

```
SQL> SELECT C.Customer_ID, C.Customer_Name,
  2         SUM(O.Total_Amount) AS Total_Purchase
  3  FROM Customer C
  4  JOIN Orders O
  5  ON C.Customer_ID = O.Customer_ID
  6  GROUP BY C.Customer_ID, C.Customer_Name;

CUSTOMER_ID CUSTOMER_NAME                                  TOTAL_PURCHASE
----------- -------------------------------------------------- --------------
        101 Arun Kumar                                               1499
        102 Priya Sharma                                             2499
        103 Rahul Raj                                                 899
        104 Divya Sri                                                3598
```

Conclusion

Thus, SQL queries were used to retrieve, filter, sort, and analyze product and customer information in the ShopSphere Online E-Commerce Database Management System. The queries demonstrate the use of SELECT, WHERE, ORDER BY, DISTINCT, filtering operators, aggregate functions, and GROUP BY for generating basic business reports.